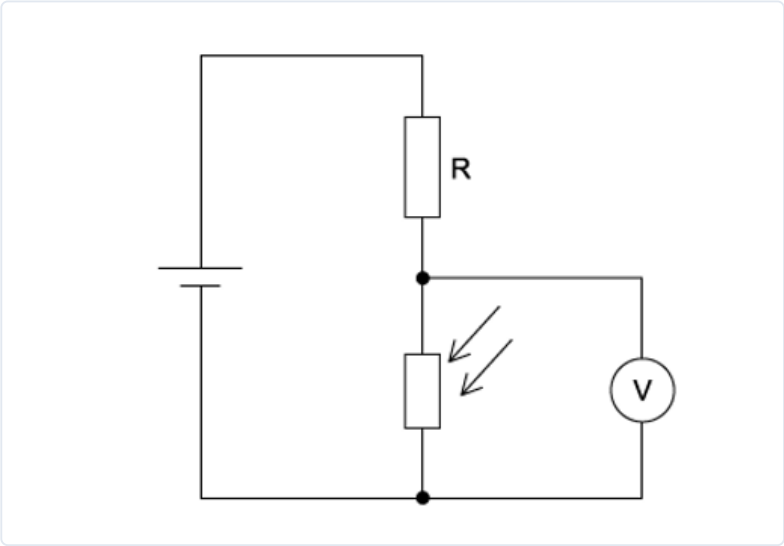
CIE AS PHYSICS (9702) • QUESTION PAPER

10.3 Potential Dividers

Syllabus Section: 10. D.C. Circuits **Paper Type:** Multiple Choice (Paper 1) **Total:** 29 Questions (29 Marks)

Question 1 Easy [1 mark]

A potential divider consists of a fixed resistor R and a light-dependent resistor (LDR)



What happens to the voltmeter reading, and why does it happen, when the intensity of light on the LDR increases?

A The voltmeter reading decreases because the LDR resistance decreases

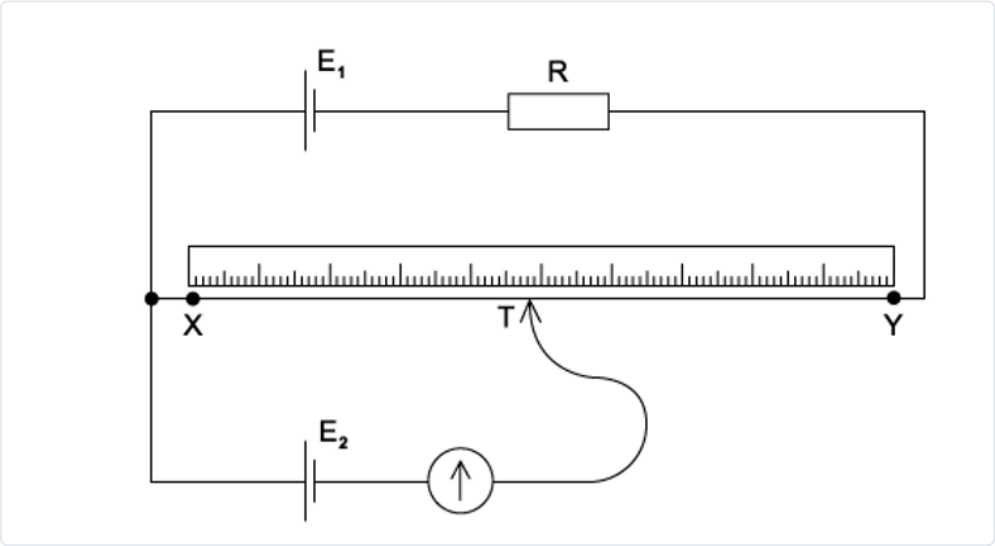
B The voltmeter reading decreases because the LDR resistance increases

C The voltmeter reading increases because the LDR resistance decreases

D the voltmeter reading increases because the LDR resistance increases

Question 2 Easy [1 mark]

The diagram shows a potentiometer circuit.



The contact T is placed on the wire and moved along the wire until the galvanometer reading is zero. The length XT is then noted.

In order to calculate the potential difference per unit length of the wire XY, which value must also be known?

A the e .m .f. of the cell E_1

B the e. m .f. of the cell E_2

C the resistance of resistor R

D the resistance of the wire XY

Question 3 Easy [1 mark]

A source of e.m.f. 9.0 mV has an internal resistance of 6.0 Ω.

It is connected across a galvanometer of resistance 30 Ω.

What is the current in the galvanometer?

A 250 μA

B 300 μA

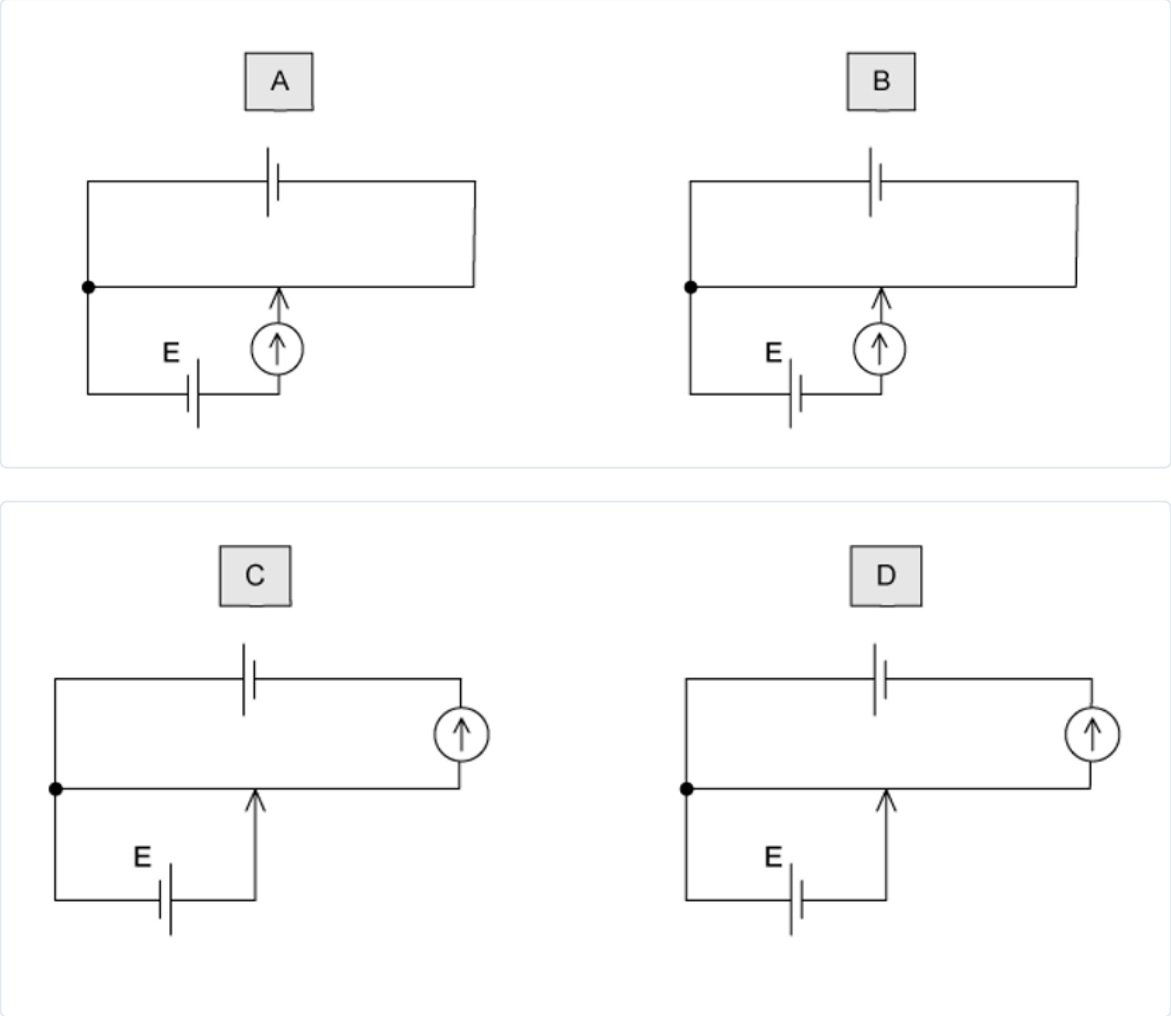
C 1.5 mA

D 2.5 mA

Question 4 Easy [1 mark]

The unknown e.m.f. *E* of a cell is to be determined using a potentiometer circuit. The balance length is to be measured when the galvanometer records a null reading.

What is the correct circuit to use ?



A

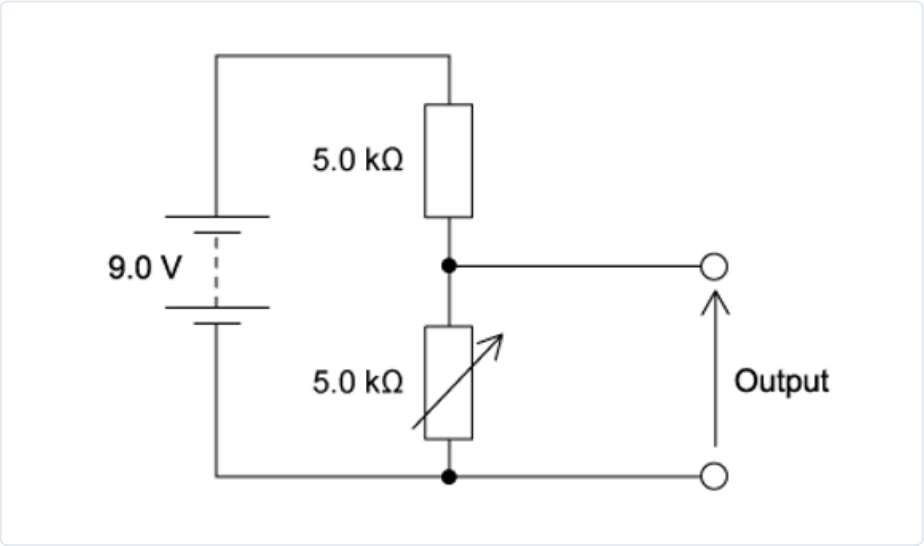
B

C

D

Question 5 Easy [1 mark]

The diagram shows a potential divider circuit designed to provide a variable output p. d.



Which gives the available range of output p. d ?

	Maximum output	Minimum output
A	3.0 V	0
B	4.5 V	0
C	9.0 V	0
D	9.0 V	4.5 V

A

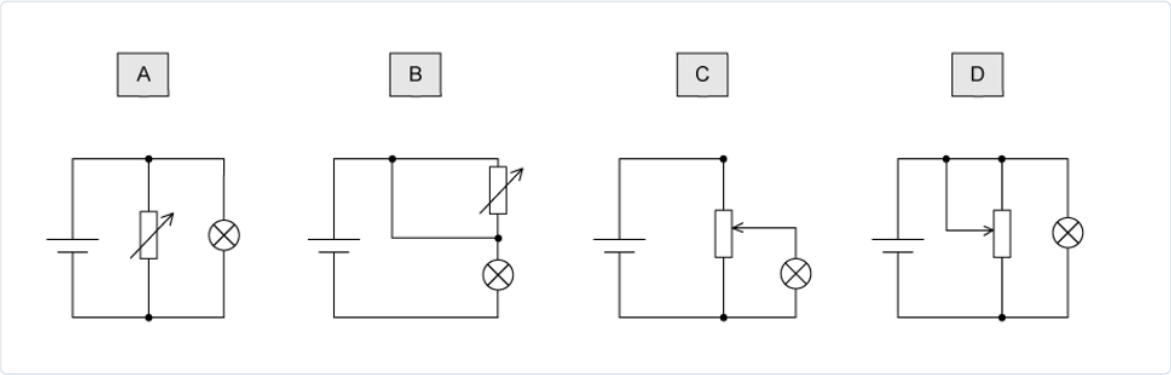
B

C

D

Question 6 Easy [1 mark]

Which diagram shows a potential divider circuit that can vary the voltage across the lamp ?



A

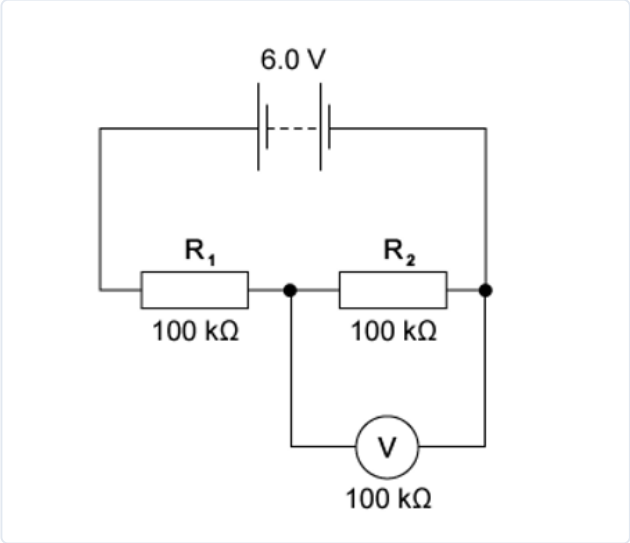
B

C

D

Question 7 Easy [1 mark]

In the circuit shown, the 6.0 V battery has negligible internal resistance. Resistors R_1 and R_2 and the voltmeter have resistance 100 kΩ.



What is the current in the resistor R_2?

A 20 μA

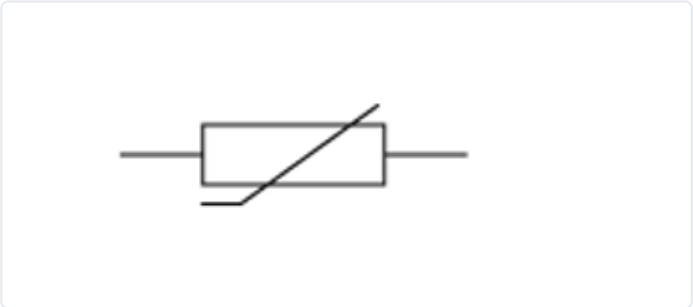
B 30 μA

C 40 μA

D 60 μA

Question 8 Easy [1 mark]

Which electrical component is represented by the following symbol ?



A a diode

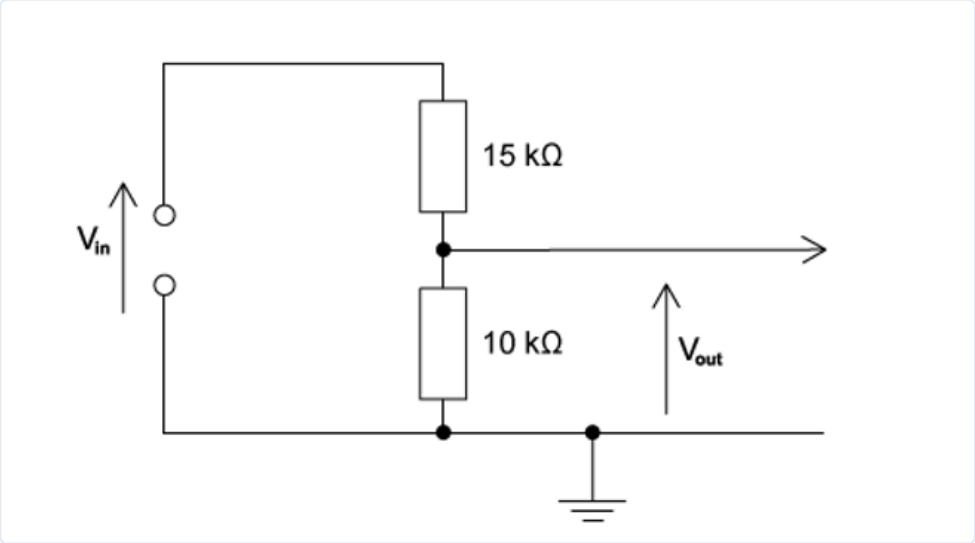
B a potentiometer

C a galvanometer

D a thermistor

Question 9 Easy [1 mark]

The circuit is designed to trigger an alarm system when the input voltage exceeds some preset value. It does this by comparing V_{out} with a fixed reference voltage, which is set at 4.8 V .



V_{out} is equal to 4.8 V.

What is the input voltage V_{in}?

A 4.8 V

B 7.2 V

C 9.6 V

D 12 V

Question 10 Easy [1 mark]

A low-voltage supply with an e. m. f. of 20 V and an internal resistance of 1.5 Ω is used to supply power to a heater of resistance 6.5 Ω in a fish tank.

What is the power supplied to the water in the fish tank ?

A 41 W

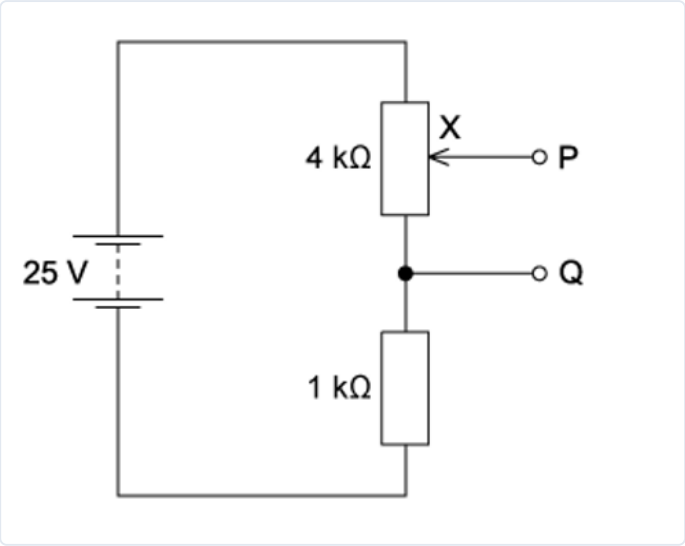
B 50 W

C 53 W

D 62 W

Question 11 Medium [1 mark]

The diagram shows a potential divider circuit which, by adjustment of the contact X, can be used to provide a variable potential difference between the terminals P and Q.



What are the limits of this potential difference?

A 0 and 5 V

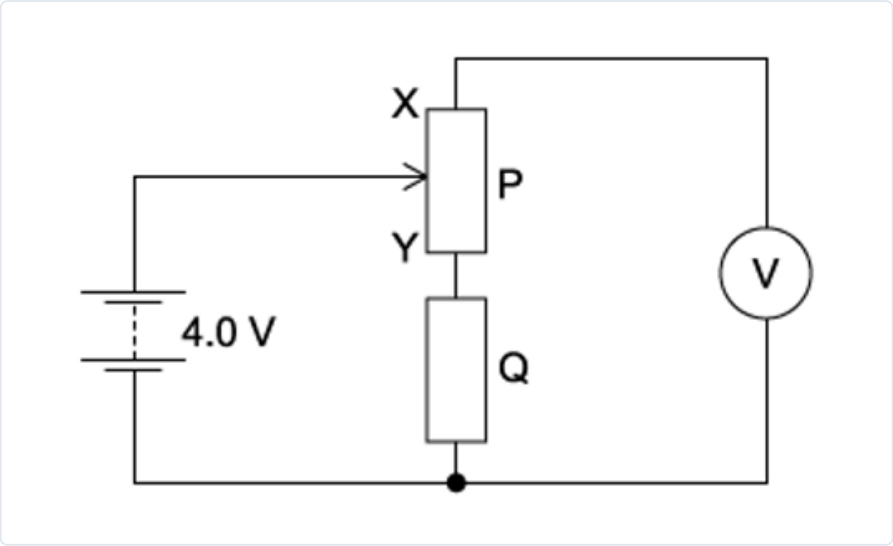
B 0 and 20 V

C 0 and 25 V

D 5 V and 25 V

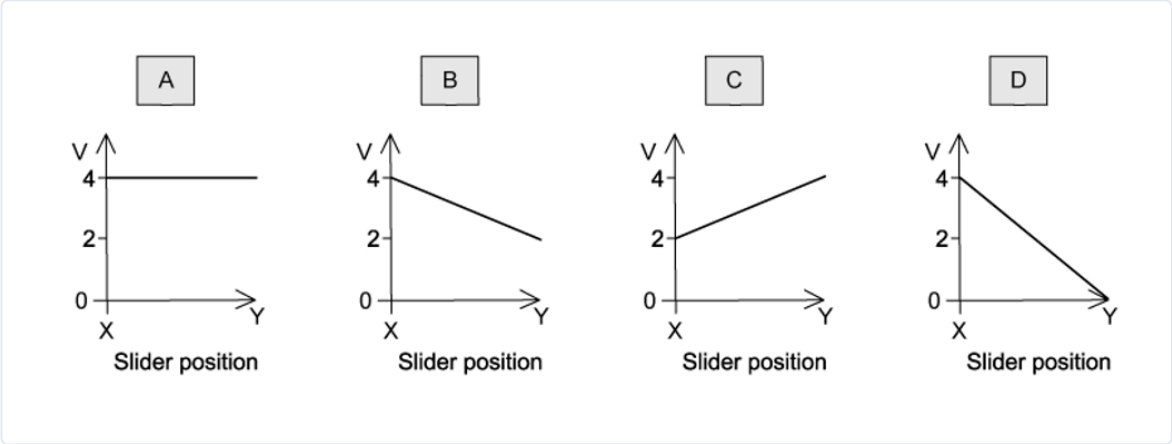
Question 12 Medium [1 mark]

In the circuit below, P is a potentiometer of total resistance 10 Ω and Q is a fixed resistor of resistance 10 Ω. The battery has an e .m f. of 4.0 V and negligible internal resistance. The voltmeter has a very high resistance .



The slider on the potentiometer is moved from X to Y, and a graph of voltmeter reading *V* is plotted against slider position.

Which graph would be obtained ?



A

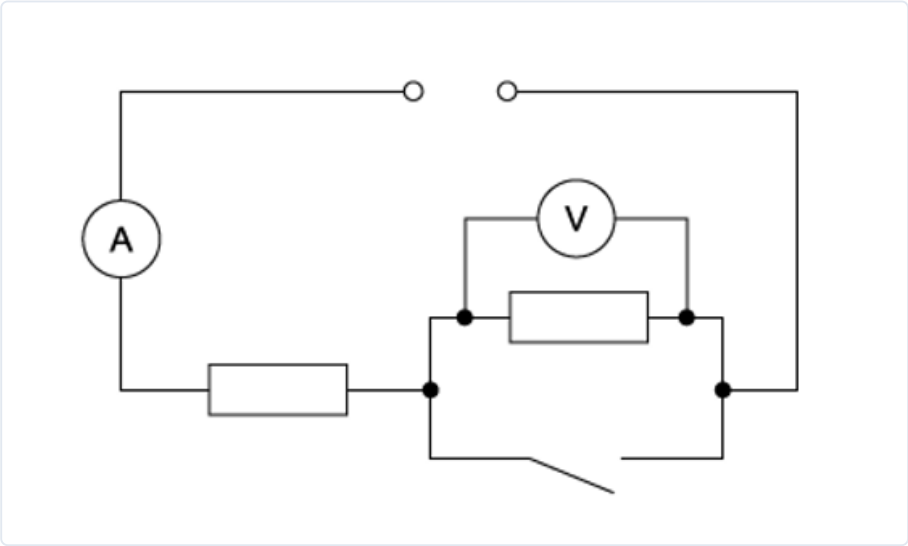
B

C

D

Question 13 Medium [1 mark]

In the circuit below, the ammeter reading is *I*, and the voltmeter reading is *V*.



When the switch is closed, which row describes what happens to *I* and *V*?

	I	*V*
A	decreases	decreases to zero
B	increases	stays the same
C	increases	decreases to zero
D	stays the same	increases

A

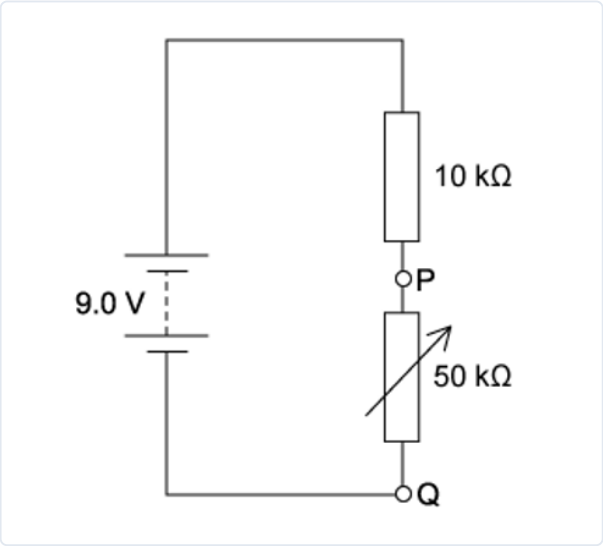
B

C

D

Question 14 Medium [1 mark]

The diagram shows a potential divider connected to a 9.0 V supply of negligible internal resistance.



What range of voltages can be obtained between P and Q?

A zero to 1.5 V

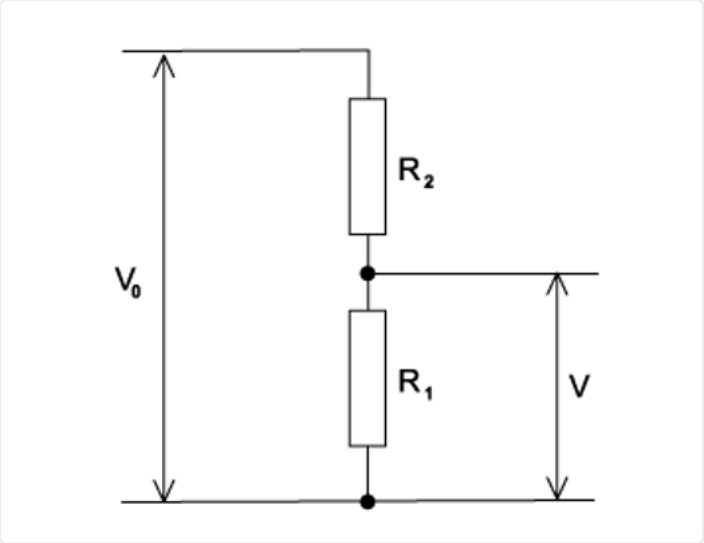
B zero to 7.5 V

C 1.5 V to 7.5 V

D 1.5 V to 9.0 V

Question 15 Medium [1 mark]

A potential divider consisting of resistors of resistance R_1 and R_2 is connected to an input potential difference of V_0 and gives an output p .d. of V.



What is the value of V ?

A $\frac{V_0 \quad R_1}{R_2}$

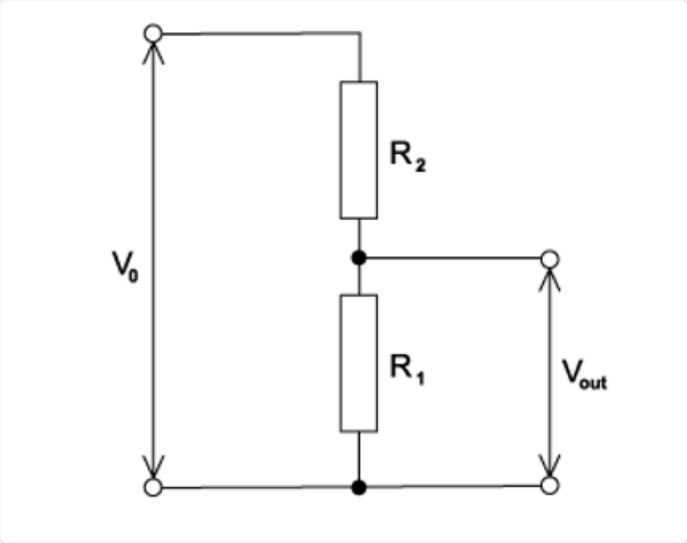
B $\frac{V_0 \quad R_1}{R_1 + R_2}$

C $\frac{V_0 \quad R_2}{R_1 + R_2}$

D $\frac{V_0 R_1 + R_2}{R_1}$

Question 16 Medium [1 mark]

A potential divider consists of resistors of resistance R_1 and R_2 connected in series across a source of potential difference V_0. The potential difference across R_1 is V_{out}.



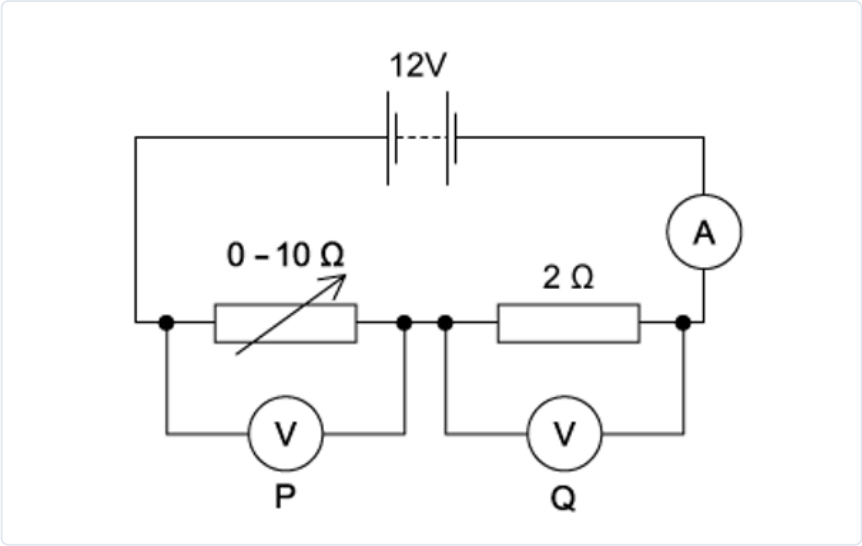
Which changes to R_1 and R_2 will increase the value of V_{out}?

	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

- **A**
- **B**
- **C**
- **D**

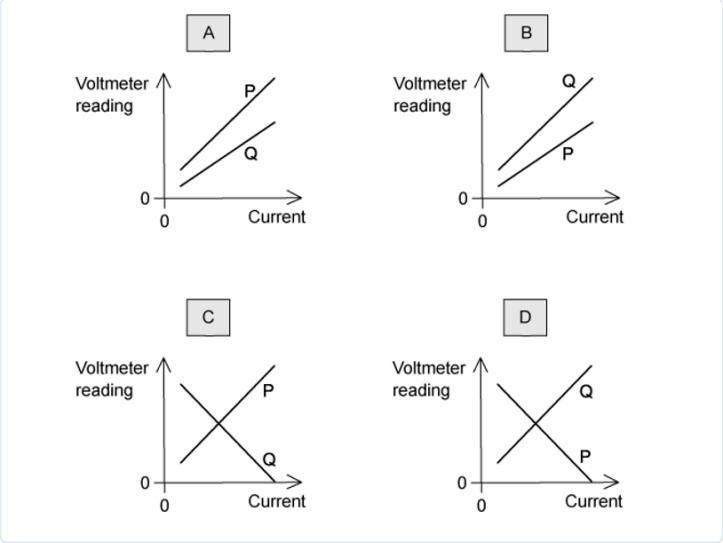
Question 17 Medium [1 mark]

A 12 V battery is in series with an ammeter, a 2 Ω fixed resistor and a 0 – 10 Ω variable resistor. High-resistance voltmeters P and Q are connected across the variable resistor and the fixed resistor, respectively, as shown.



The resistance of the variable resistor is changed from its maximum value to zero.

Which graph shows the variation with current of the voltmeter readings?



A

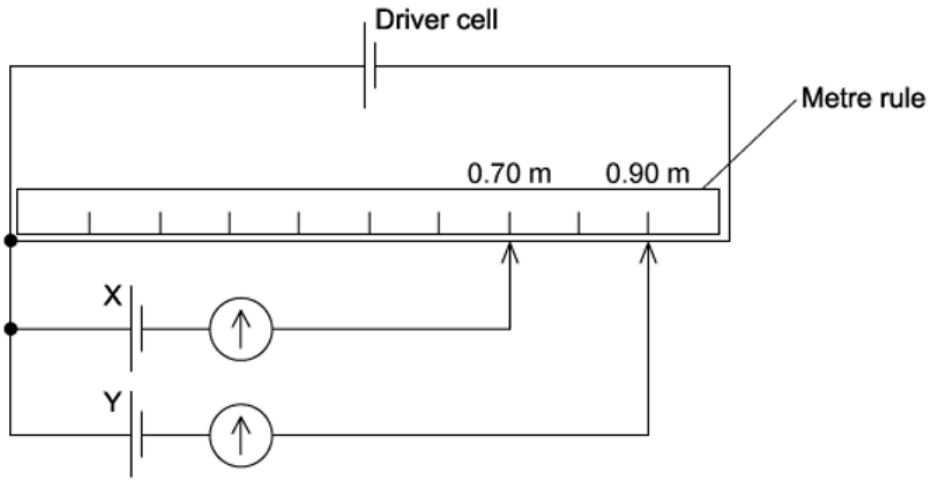
B

C

D

Question 18 Medium [1 mark]

A potentiometer is used as shown to compare the e.m.f. of two cells.



The balance points for cells X and Y are 0.70 m and 0.90 m respectively.

If the e.m.f. of cell X is 1.1 V, what is the e.m.f. of cell Y ?

A 0.69 V

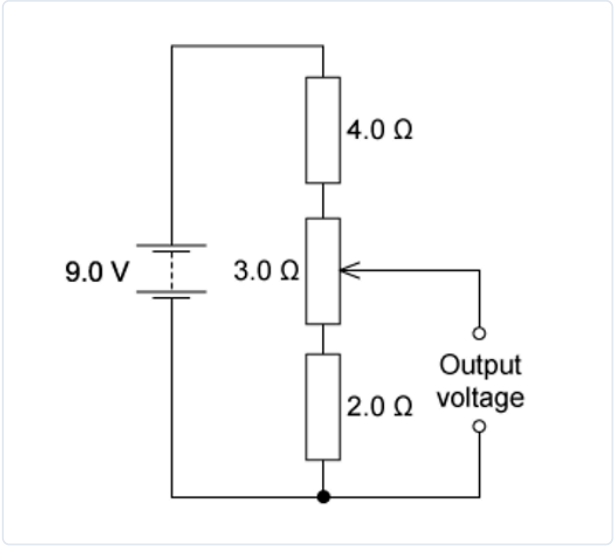
B 0.86 V

C 0.99 V

D 1.4 V

Question 19 Medium [1 mark]

A potential divider circuit consists of fixed resistors of resistance 2.0 Ω and 4.0 Ω connected in series with a 3.0 Ω resistor fitted with a sliding contact. These are connected across a battery of e.m.f. 9.0 V and zero internal resistance, as shown.



What are the maximum and minimum output voltages of this potential divider circuit?

	Maximum voltage / V	Minimum voltage / V
A	4.0	2.0
B	5.0	2.0
C	9.0	0
D	9.0	2.0

A

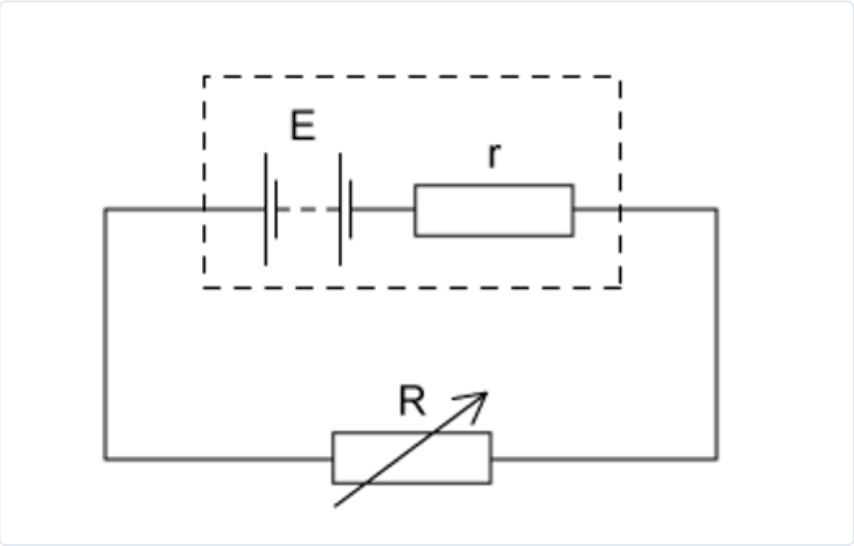
B

C

D

Question 20 Medium [1 mark]

A battery with e.m.f. *E* and internal resistance *r* is connected in series with an external variable resistor.



The value of the external resistance *R* is slowly increased from zero.

Which statement is correct ? (Ignore any temperature effects.)

A the potential difference across the external resistance decreases

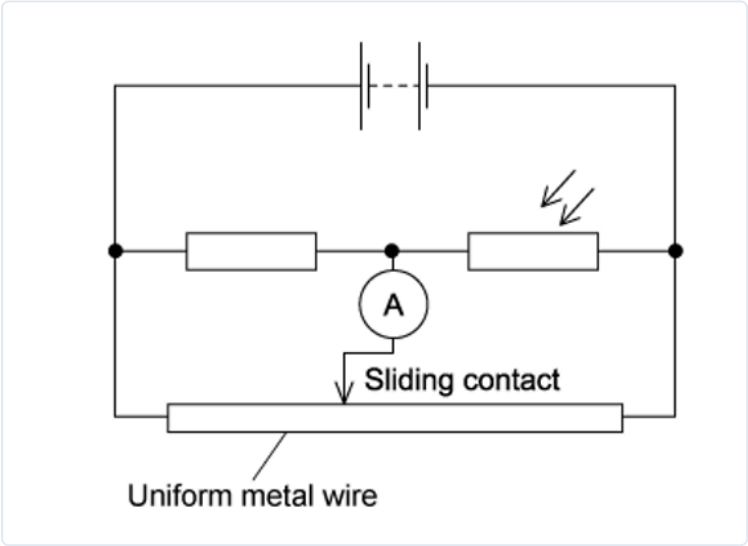
B the potential difference across the internal resistance increases

C the power dissipated in *r* increases and then decreases

D the power dissipated in *R* increases and then decreases

Question 21 Hard [1 mark]

In the potentiometer circuit shown, the reading on the ammeter is zero.



The light-dependent resistor (LDR) is then covered up, and the ammeter gives a non-zero reading.

Which change could return the ammeter reading to zero?

A decrease the supply voltage

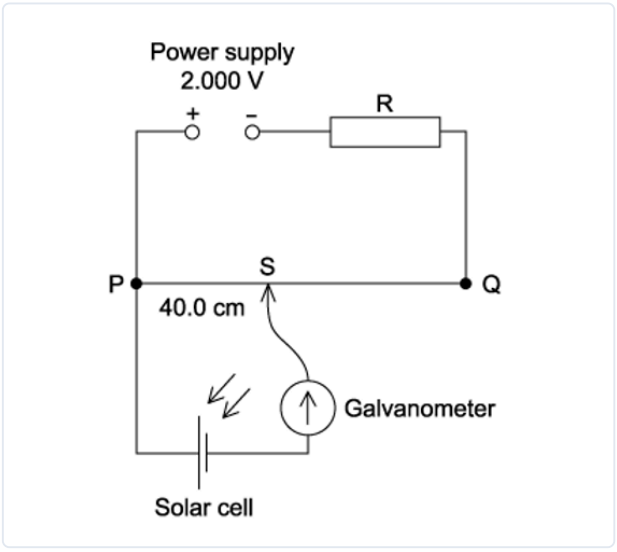
B increase the supply voltage

C move the sliding contact to the left

D move the sliding contact to the right

Question 22 Hard [1 mark]

A power supply and a solar cell are compared using the potentiometer circuit shown.



The e.m.f. produced by the solar cell is measured on the potentiometer.

The potentiometer wire PQ is 100.0 cm long and has a resistance of 5.00 Ω. The power supply has an e. m .f. of 2.000 V, and the solar cell has an e .m. f. of 5.00 mV.

Which resistance *R* must be used so that the galvanometer reads zero when PS = 40.0 cm ?

A 395 Ω

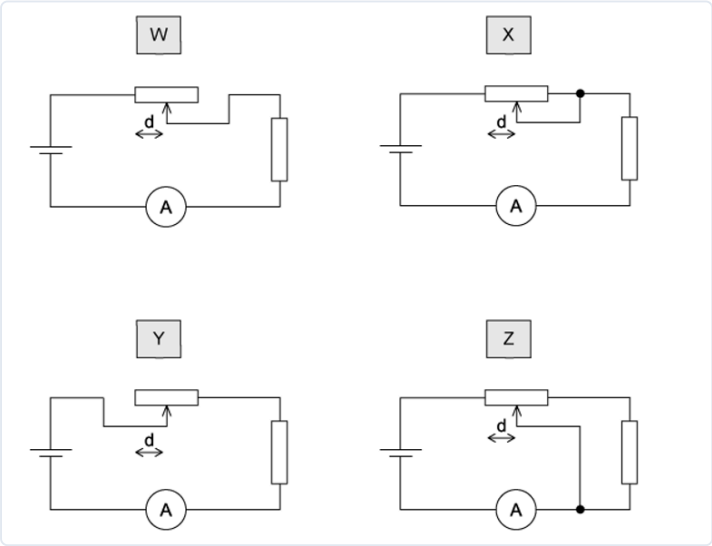
B 795 Ω

C 995 Ω

D 1055 Ω

Question 23 Hard [1 mark]

The diagrams show the same cell, ammeter, potentiometer and fixed resistor connected in different ways.



The distance *d* between the sliding contact and a particular end of the potentiometer is varied. The current measured is then plotted against the distance *d*.

For which two circuits will the graphs be identical ?

A W and X

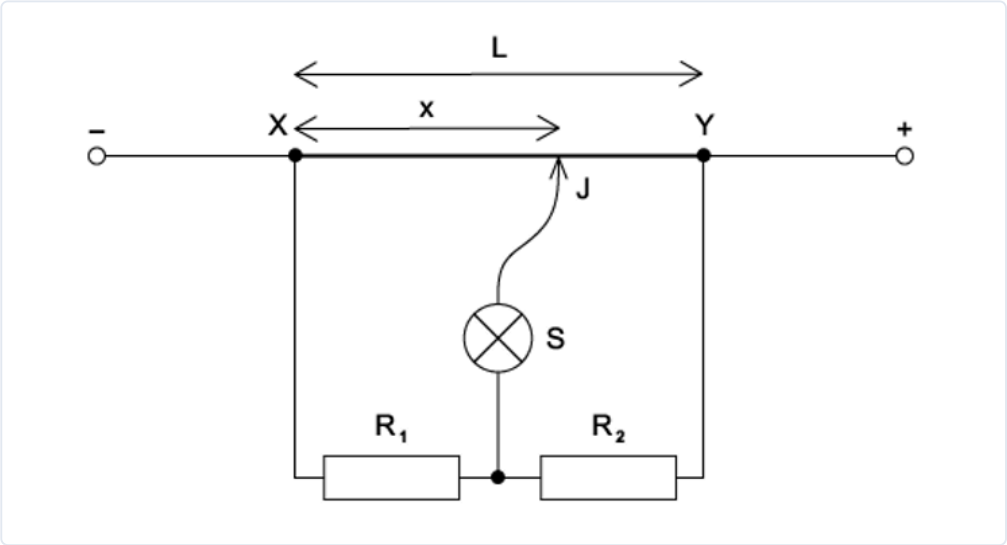
B W and Y

C X and Y

D Y and Z

Question 24 Hard [1 mark]

In the circuit shown, XY is a length *L* of uniform resistance wire. R_1 and R_2 are unknown resistors. J is a sliding contact that joins the junction of R_1 and R_2 to points on XY through a small signal lamp S.



To determine the ratio $\frac{V_1}{V_2}$ of the potential differences across R_1 and R_2, a point is found on XY at which the lamp is off. This point is at a distance *x* from *X*.

What is the value of the ratio $\frac{V_1}{V_2}$?

A $\frac{L}{x}$

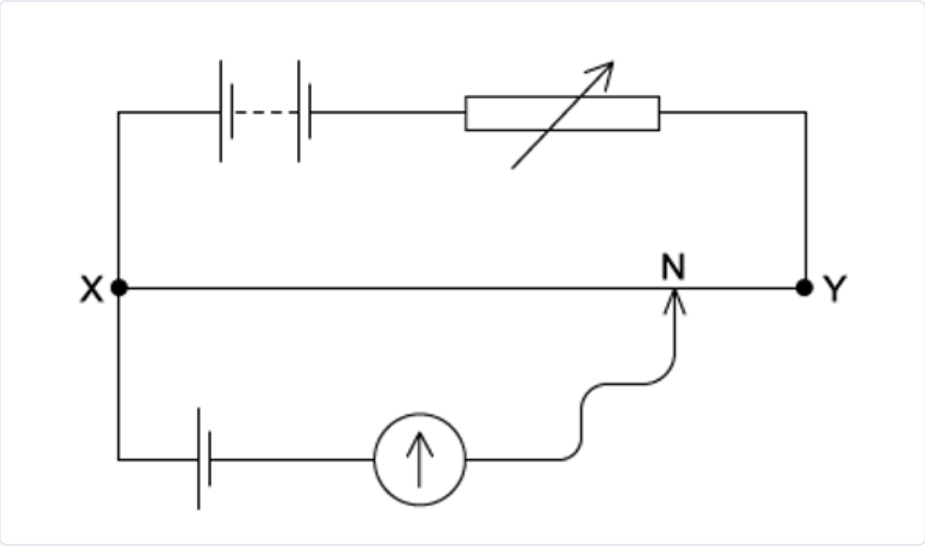
B $\frac{x}{L}$

C $\frac{L-x}{x}$

D $\frac{x}{L-x}$

Question 25 Hard [1 mark]

In the potentiometer circuit below, the moveable contact is placed at N on the bare wire XY, such that the galvanometer shows zero deflection.



The resistance of the variable resistor is now increased.

What is the effect of this increase on the potential difference across the wire XY and on the position of the moveable contact for zero deflection ?

	Potential difference across XY	Position of moveable contact
A	increases	nearer to X
B	increases	nearer to Y
C	decreases	nearer to X
D	decreases	nearer to Y

A

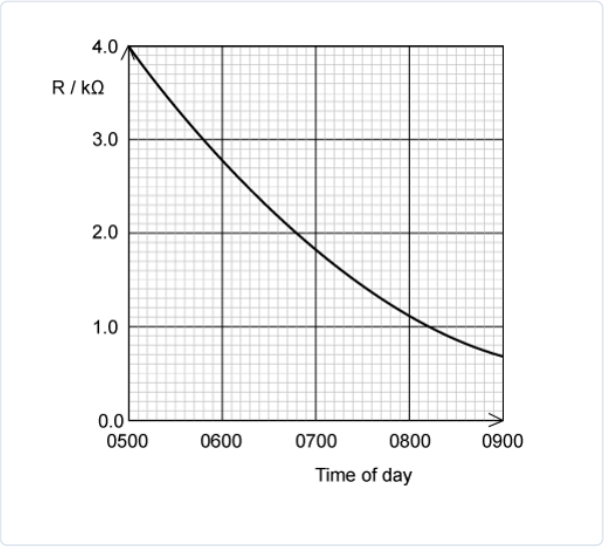
B

C

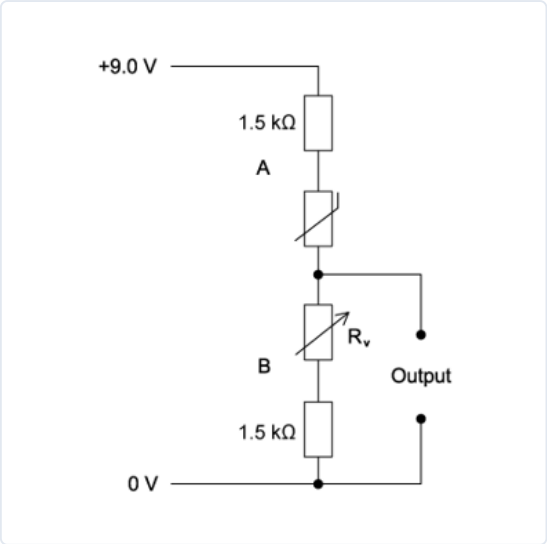
D

Question 26 Hard [1 mark]

The graph below shows how resistance *R* of a thermistor varies during part of a day.



The thermistor is connected to the potential divider circuit as shown.



At 0730, an output of 6.0 V is required.

What value of the variable resistor should be set at this time?

A 1.5 kΩ

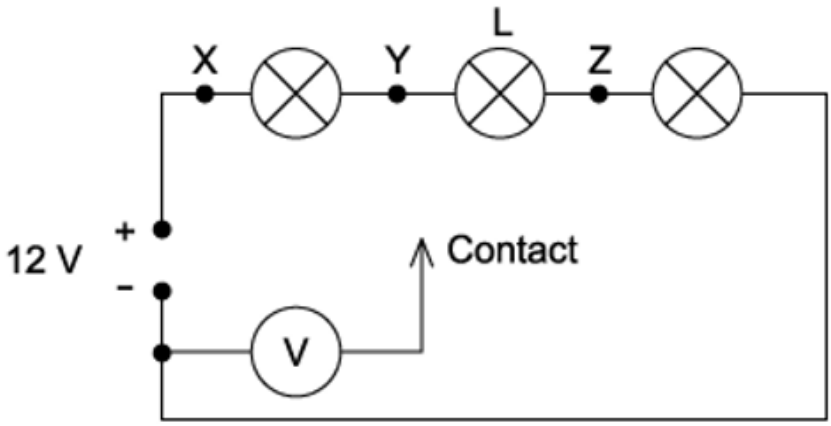
B 3.0 kΩ

C 4.5 kΩ

D 6.0 kΩ

Question 27 Hard [1 mark]

The contact is connected one at a time to points X, Y and Z to test the circuit. The lamps do not light up because lamp L is blown.



Which of the following are the readings of the voltmeter when the contact is connected to each of the points?

	X	Y	Z
A	12 V	8 V	4 V
B	8 V	8 V	0 V
C	12 V	12 V	0 V
D	8 V	12 V	4 V

A

B

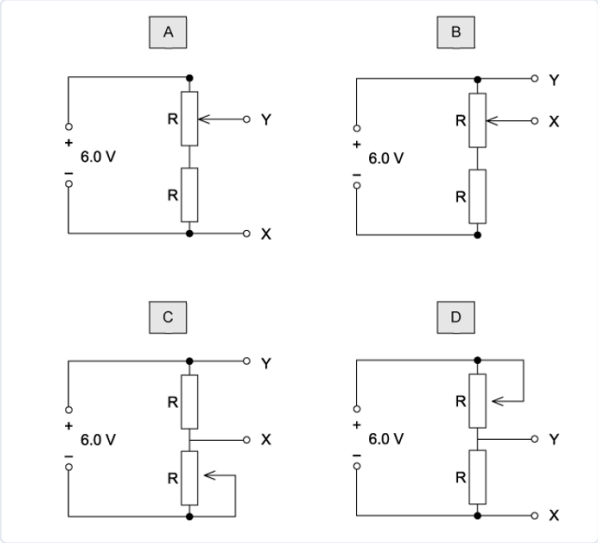
C

D

Question 28 Hard [1 mark]

A potential divider has a constant supply of 6.0 V, as shown in the diagrams.

Which circuit will provide a potential difference between X and Y that can be varied between zero and 3.0 V ?



A

B

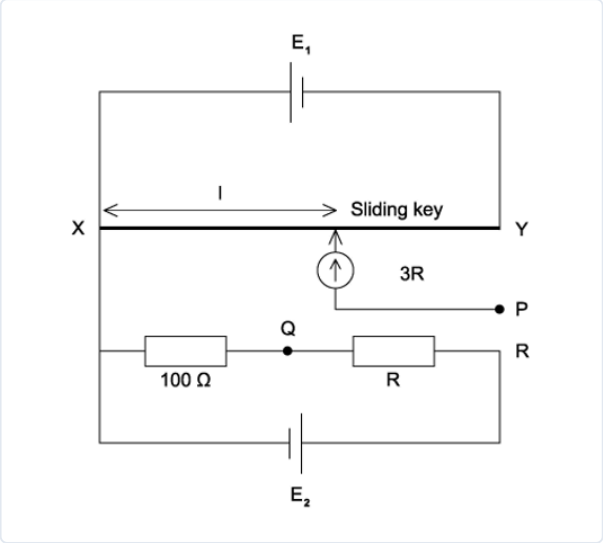
C

D

Question 29 Hard [1 mark]

The resistance *R* of an unknown resistor is compared with a fixed 100 Ω resistor using the circuit as shown in the diagram.

The balance length *l* is 30.0 cm and 48.3 cm when P is connected to Q and R, respectively.



The length of the wire XY is 100 cm and has a resistance of 40 Ω. The two cells E_1 and E_2, have negligible internal resistances.

What is the value of resistance *R* ?

A 38 Ω

B 47 Ω

C 61 Ω

D 161 Ω